Evaluating Limits

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Direct substitution and algebraic techniques

Evaluate each limit:

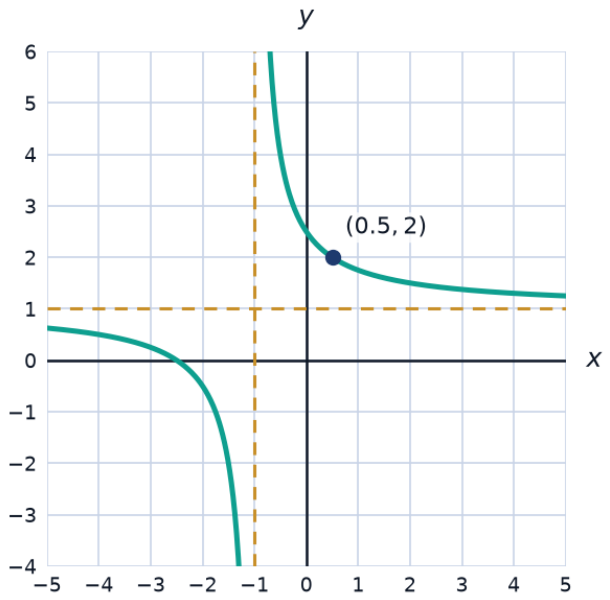
a $\lim_{x \to 1} (3x^2 - 2x + 5)$

b $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$

c $\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$

2 [Foundational] • Target: Direct substitution and algebraic techniques

Evaluate $\lim_{x \to 3} \frac{x^2 - x - 6}{x^2 - 9}$.



3 [Foundational] • Target: Trigonometric and special limits

Using $\lim_{x \to 0} \frac{\sin x}{x} = 1$, evaluate:

a $\lim_{x \to 0} \frac{\sin 5x}{x}$

b $\lim_{x \to 0} \frac{1 - \cos x}{x}$

c $\lim_{x \to 0} \frac{\sin 3x}{\sin 5x}$

4 [Foundational] • Target: Trigonometric and special limits

Evaluate $\lim_{x \to \infty} \frac{5x^3 - 2x + 1}{2x^3 + 7}$.

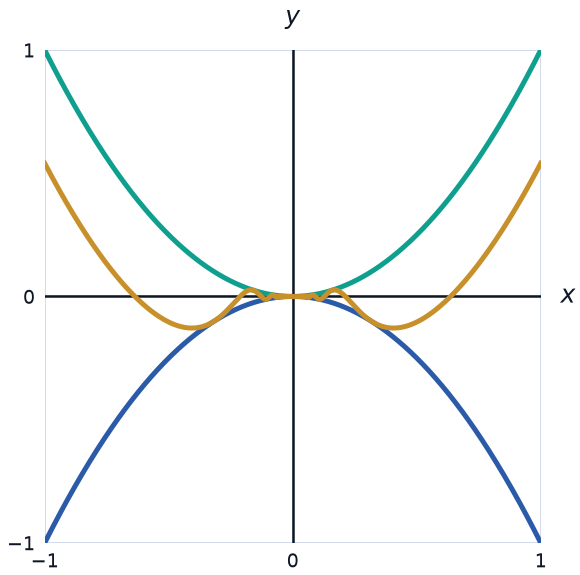
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5 [Foundational] • Target: Limits that fail to exist and the squeeze theorem

Explain why $\lim_{x \to 0} \sin\left(\frac{1}{x}\right)$ does not exist.

6 [Foundational] • Target: Limits that fail to exist and the squeeze theorem

Use the squeeze theorem to evaluate $\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$.



7 [Foundational] • Target: Limits with absolute value and piecewise functions

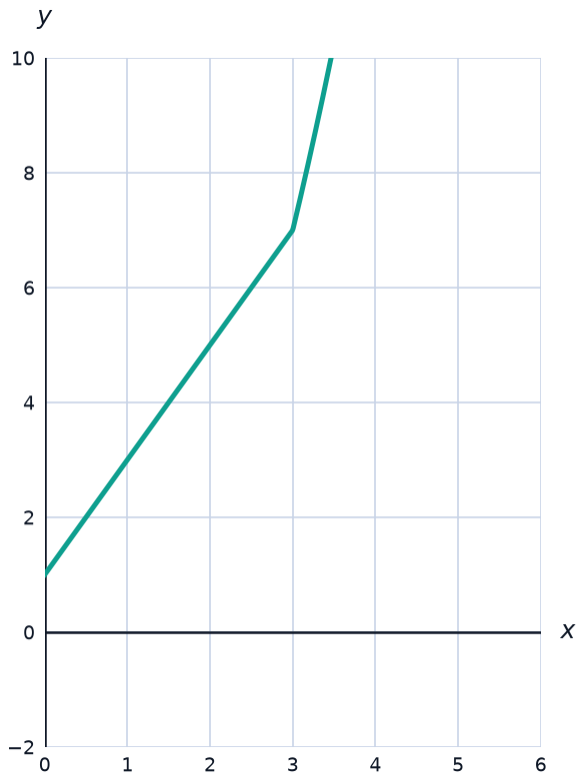
Evaluate $\lim_{x \to 3} |x - 3| \cdot \frac{1}{x}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Limits with absolute value and piecewise functions**

For $f(x) = 2x + 1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \geq 3$, evaluate $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$, and state whether $\lim_{x \to 3} f(x)$ exists.

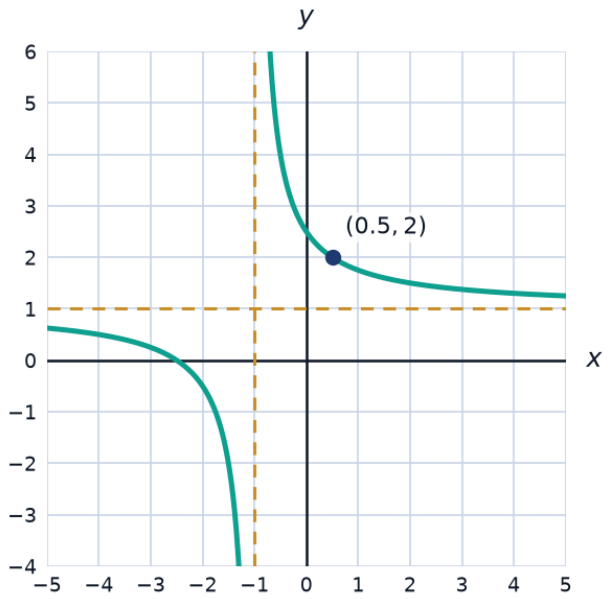


9 **[Intermediate]** • **Target: More algebraic limit techniques**

Evaluate $\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$ by factoring with the difference-of-cubes formula.

10 **[Intermediate]** • **Target: More algebraic limit techniques**

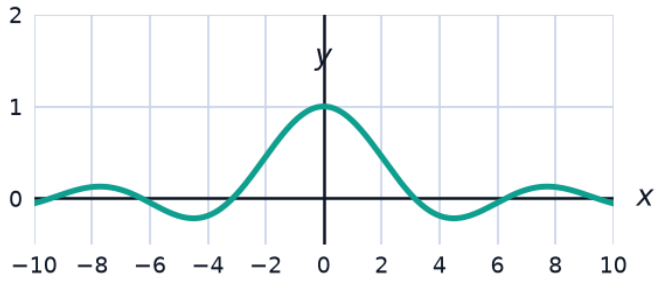
Evaluate $\lim_{h \to 0} \frac{(2 + h)^2 - 4}{h}$.



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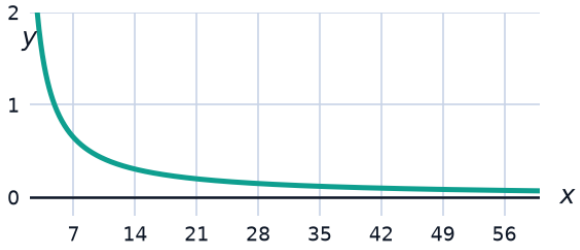
11 [Intermediate] • Target: Visualizing a limit numerically and graphically

The function $f(x) = \frac{\sin x}{x}$ (undefined at $x = 0$) is graphed below. Use the graph to estimate $\lim_{x \to 0} f(x)$.



12 [Intermediate] • Target: Limits at infinity with different degree relationships

Evaluate each limit at infinity, based on the relative degrees of numerator and denominator:



a $\lim_{x \to \infty} \frac{4x + 3}{x^2 - 1}$ (lower degree numerator)

b $\lim_{x \to \infty} \frac{x^3 - 1}{2x^2 + 5}$ (higher degree numerator)

13 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x \to 9} \frac{x - 9}{\sqrt{x} - 3}$ by rationalizing the denominator.

14 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x \to \infty} \left(\sqrt{x^2 + 3x} - x\right)$ by rationalizing.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

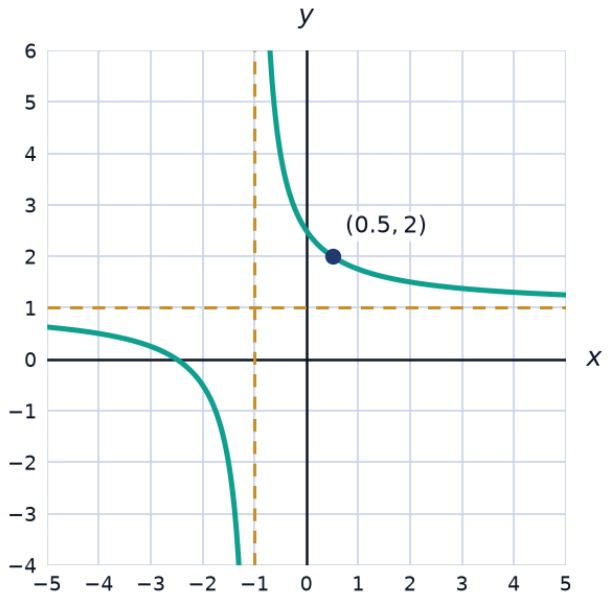
15 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 1.01 Precalculus Review for Calculus

For $f(x) = kx + 1$ if $x \le 2$, and $f(x) = x^2 - 3$ if $x > 2$, find k so that $\lim_{x \to 2} f(x)$ exists.

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16 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Evaluate $\lim_{x \to 1} \frac{x^4 - 1}{x - 1}$ by factoring the numerator as a difference of squares twice.



17 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 11.01 Sequences and Series

Evaluate $\lim_{x \to -2} \frac{x^3 + 8}{x + 2}$ using the sum-of-cubes factoring pattern.

18 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 12.01 Convergence Tests

Evaluate $\lim_{x \to 0} \frac{(3 + x)^2 - 9}{x}$.

19 [Challenging] • Target: More algebraic limit techniques • Cross-Lesson: 12.02 Power Series and Taylor Series

The figure shows the graph of a function $f(x)$. What is the value of $\lim_{x \to 2} f(x)$?

20 [Challenging] • Target: Visualizing a limit numerically and graphically • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

$\frac{d}{dx}\left[(\ln x)^{\ln x}\right] =$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Direct Substitution and Algebraic Techniques

Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3}$.

a. 3/2

b. 0

c. 1

d. Does not exist

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22 [Foundational] • Target: Direct Substitution and Algebraic Techniques

Evaluate $\lim_{x \to 0} \frac{\sqrt{x+4}-2}{x}$.

a. 1/4

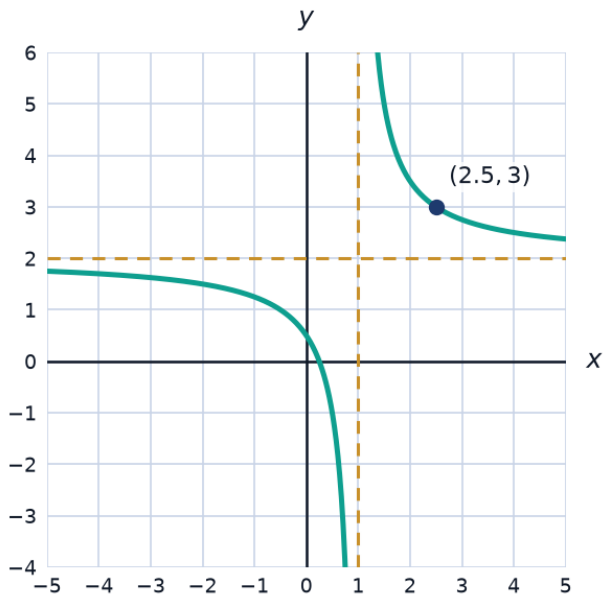
b. 1/2

c. 0

d. 1/8

23 [Foundational] • Target: Direct Substitution and Algebraic Techniques

Evaluate $\lim_{x \to 1} \frac{\frac{1}{x+1}-\frac{1}{2}}{x-1}$.



a. $-1/4$

b. 1/4

c. $-1/2$

d. 0

24 [Foundational] • Target: Trigonometric Limits and the Squeeze Theorem

Evaluate $\lim_{x \to 0} \frac{\sin(5x)}{3x}$.

a. 5/3

b. 3/5

c. 1

d. 0

25 [Foundational] • Target: Trigonometric Limits and the Squeeze Theorem

Evaluate $\lim_{x \to 0} \frac{1-\cos(x)}{x^2}$.

a. 1/2

b. 0

c. 1

d. Does not exist

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26 [Foundational] • Target: Trigonometric Limits and the Squeeze Theorem

If $1 - \frac{x^2}{6} \le g(x) \le 1$ for all x near 0, find $\lim_{x \to 0} g(x)$.

a. 1

b. 0

c. 5/6

d. Cannot be determined

27 [Foundational] • Target: Infinite Limits and Limits at Infinity

Evaluate $\lim_{x \to \infty} \frac{4x^3 - 2x + 7}{5x^3 + 3x^2 - 1}$.

a. 4/5

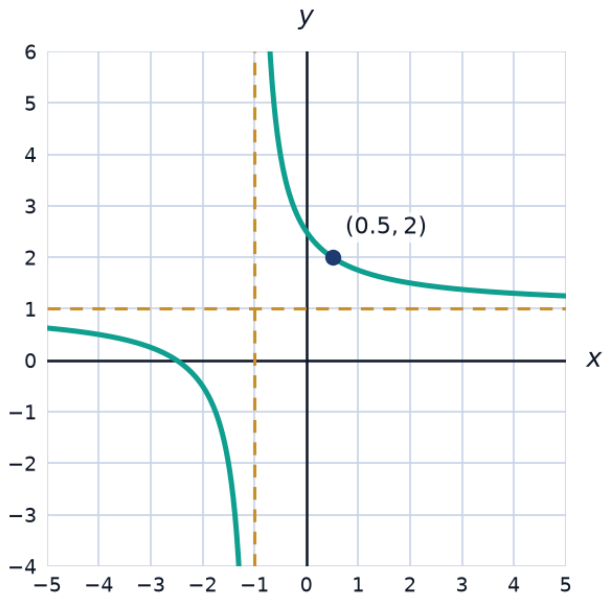
b. 0

c. ∞

d. $7/(-1)$

28 [Foundational] • Target: Infinite Limits and Limits at Infinity

Evaluate $\lim_{x \to -\infty} \frac{\sqrt{9x^2 + 4}}{2x - 1}$.



a. $-3/2$

b. 3/2

c. 9/2

d. $-9/2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Infinite Limits and Limits at Infinity

Find $\lim_{x\to 2^+} \frac{3x-1}{x-2}$ and $\lim_{x\to 2^-} \frac{3x-1}{x-2}$.

a. $+\infty$ and $-\infty$

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 5 and 5

30 [Intermediate] • Target: Piecewise and One-Sided Limits

Let $f(x)$ be defined as $x^2 + 2$ for $x < 1$ and $4 - x$ for $x \geq 1$. Evaluate $\lim_{x\to 1} f(x)$.

a. 3

b. 1

c. 2

d. Does not exist

31 [Intermediate] • Target: Piecewise and One-Sided Limits

Evaluate $\lim_{x\to 0^+} \frac{|x|}{x}$ and $\lim_{x\to 0^-} \frac{|x|}{x}$.

a. 1 and -1

b. 1 and 1

c. -1 and -1

d. 0 and 0

32 [Intermediate] • Target: Piecewise and One-Sided Limits

Find the value of k such that $\lim_{x\to 2} g(x)$ exists, where $g(x)$ is $kx - 3$ for $x \leq 2$ and $x^2 + 1$ for $x > 2$.

a. 4

b. 2

c. 5

d. 3

33 [Intermediate] • Target: Infinite Limits and Limits at Infinity

One-sided limit: $f(x) = |x|/x$. $\lim_{x\to 0^-} f(x) =$

a. 1

b. -1

c. 0

d. DNE

34 [Intermediate] • Target: Piecewise and One-Sided Limits

Evaluate the one-sided limit $\lim_{x\to -1^-} \frac{|x+1|}{x+1}$.

a. 0

b. ∞

c. -1

d. 1

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35 [Intermediate] • Target: Direct substitution and algebraic techniques

Evaluate the one-sided limit $\lim_{x\to -2^-} \frac{|x+2|}{x+2}$.

a. 0

b. ∞

c. 1

d. -1

36 [Intermediate] • Target: Trigonometric and special limits

Evaluate the one-sided limit $\lim_{x\to -3^-} \frac{|x+3|}{x+3}$.

a. 1

b. ∞

c. -1

d. 0

37 [Intermediate] • Target: Limits that fail to exist and the squeeze theorem

Evaluate the one-sided limit $\lim_{x\to -4^-} \frac{|x+4|}{x+4}$.

a. 0

b. 1

c. -1

d. ∞

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits with absolute value and piecewise functions • Cross-Lesson: 12.01 Convergence Tests

Evaluate the one-sided limit $\lim_{x\to 3^-} \frac{|x-3|}{x-3}$.

a. 0

b. ∞

c. -1

d. 1

39 [Challenging] • Target: More algebraic limit techniques • Cross-Lesson: 12.02 Power Series and Taylor Series

Evaluate the one-sided limit $\lim_{x\to 2^+} \frac{|x-2|}{x-2}$.

a. 1

b. -1

c. 0

d. ∞

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40 [Challenging] • Target: Visualizing a limit numerically and graphically • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Evaluate the one-sided limit $\lim_{x \to 4^+} \frac{|x-4|}{x-4}$.

a. 1

b. ∞

c. -1

d. 0

41 [Challenging] • Target: Limits at infinity with different degree relationships • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Evaluate the one-sided limit $\lim_{x \to 5^+} \frac{|x-5|}{x-5}$.

a. 0

b. -1

c. ∞

d. 1

42 [Challenging] • Target: Limits requiring conjugate multiplication • Cross-Lesson: 15.01 Double Integrals

Evaluate the one-sided limit $\lim_{x \to 1^+} \frac{|x-1|}{x-1}$.

a. 0

b. 1

c. ∞

d. -1

43 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Evaluate the one-sided limit $\lim_{x \to 6^+} \frac{|x-6|}{x-6}$.

a. 1

b. 0

c. -1

d. ∞

44 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 2.02 Continuity

Consider the function $f(x)$ whose graph is given below. The function is defined piecewise. For $x \leq 2$, $f(x) = ax + b$. For $x > 2$, $f(x) = cx + d$. The graph of $f(x)$ passes through the point $(0, 3)$. The limit of $f(x)$ as x approaches 2 exists. Find the value of a.

a. -1

b. 1

c. 2

d. -2

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45 [Challenging] • Target: Direct Substitution and Algebraic Techniques • Cross-Lesson: 3.01 The Derivative from First Principles

Squeeze theorem: $-x^2 \le f(x) \le x^2$ near $x = 0$ implies $\lim_{x \to 0} f(x) =$

a. 0

b. ± 1

c. DNE

d. 1

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